

Functions

From "formula" to "function": a rule is not enough — you must say where it starts and where it lands.

Session	Date	Today
3 of 15	Wed 9 Sep 2026	Domain, codomain, undoing

Where session 3 sits

1 · Course overview and school-maths refresher

2 · Sets and set operations

3 · Functions

4 · Function families

5 · Theorem proving

6 · Mathematical induction

7 · Elementary combinatorics and binomial theorem

8 · Midterm

9 · GCD and divisibility

10 · Modular arithmetic

11 · Vectors

12 · Polynomials

13 · Complex numbers

14 · Course recap and exam preparation

15 · Final exam

TODAY

Domain and codomain, injective/surjective/bijective, composition and inverses, and the unit circle.

Yesterday gave us the containers; today we move between them

- Sets, and $\in$ against $\subseteq$
- $\cup$, $\cap$, $\setminus$, complement, $\times$
- Ordered pairs, so $A \times B$ makes sense
- Proof by double inclusion

TODAY'S QUESTION

You have written x^2 a thousand times. Is x^2 a function?

And if it is — can you run it backwards?

By the end of today you can...

- Define a function as a rule with a stated domain and codomain, and tell codomain from image
- Decide and **prove** injectivity, surjectivity, bijectivity
- Compose functions, and construct the inverse of a bijection
- Define sine, cosine and tangent on the unit circle, and read off period, range and the identities

How today runs

Two concepts, four group activities, one exit ticket.
You will be at the board.

Objective 2 is the assessed one: a verdict without a proof scores nothing.

A function is three things, not one

DEFINITION A function $f : A \rightarrow B$ is three things at once:

- a **domain** A
- a **codomain** B
- a rule giving **each** $a \in A$ **exactly one** $f(a) \in B$

Its **graph** is a subset of $A \times B$, with each a occurring in exactly one pair.

IS $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$. Every real has exactly one square, and it is real.

IS NOT

$g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = 1/x$: nothing at $x = 0$. *Not total.*

The relation $y^2 = x$ on $\mathbb{R}$: two possible y values when $x = 4$, and none when $x = -1$. *Not a function.*

The codomain is declared; the image is computed

DEFINITION

$$f(A) = \{ f(a) : a \in A \} \subseteq B$$

$$f^{-1}(S) = \{ a \in A : f(a) \in S \}$$

The first is the **image**, the second the **preimage** of $S \subseteq B$.

WORKED

$$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2.$$

Codomain $\mathbb{R}$, image $[0, \infty)$. So -1 sits in the codomain but never in the image.

$$f^{-1}(\{4\}) = \{-2, 2\}, \text{ while } f^{-1}([-1, 0)) = \emptyset.$$

NOTATION WARNING $f^{-1}(S)$ takes a **set** and returns a **set**. It exists for every function. It is not the inverse function f^{-1} we meet after the break, which needs a bijection.

Which of the three ingredients is missing?

ACTIVITY 1 GROUPS OF THREE

12 MIN

For each line: is it a function? If not, name **exactly** which requirement fails, and give the input that breaks it.

1. $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(n) = n/2$

2. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \sqrt{x}$

3. $f : \mathbb{Q} \rightarrow \mathbb{Z}, f(p/q) = p$

4. $f : \mathbb{R} \rightarrow [0, \infty), f(x) = x^2$

5. $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by the condition $f(x)^2 = x$

Share: each group takes one line and names the failure – *not total, not single-valued, lands outside the codomain, or fine.*

Name the requirement that fails

RULE	VERDICT	THE BREAKING INPUT
$1 \cdot \mathbb{Z} \rightarrow \mathbb{Z}, n/2$	Leaves the codomain	$f(3) = 1.5 \notin \mathbb{Z}.$
$2 \cdot \mathbb{R} \rightarrow \mathbb{R}, \sqrt{x}$	Not total	No value at $x = -1.$
$3 \cdot \mathbb{Q} \rightarrow \mathbb{Z}, p/q \mapsto p$	Not well defined	$\frac{1}{2} = \frac{2}{4}$, yet the rule returns 1 and 2.
$4 \cdot \mathbb{R} \rightarrow [0, \infty), x^2$	A function	Total, single-valued, lands in $[0, \infty).$
$5 \cdot \mathbb{R} \rightarrow \mathbb{R}, f(x)^2 = x$	No such function	At $x = -1$ we would need $f(-1)^2 = -1.$

LINE 3 IS THE NEW IDEA The input is a *rational number*, not a pair of integers. A rule using p and q must agree on every fraction naming that number.

Injective, surjective, bijective – three questions about hitting

DEFINITION

Injective: $f(a_1) = f(a_2) \implies a_1 = a_2$. Nothing in B is hit twice.

Surjective: $\forall b \in B \exists a \in A$ with $f(a) = b$. Nothing in B is missed.

Bijective: both. Every b hit exactly once.

SAME RULE, OPPOSITE ANSWERS

$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$: injective fails at $f(-2) = f(2)$; surjective fails at -1 .

$f : [0, \infty) \rightarrow [0, \infty), f(x) = x^2$: **both** hold. Nothing about the rule changed.

Each of the four claims has one fixed opening move

THE FOUR TEMPLATES

Injective: assume $f(a_1) = f(a_2)$, derive $a_1 = a_2$.

Not injective: exhibit one pair $a_1 \neq a_2$ with equal images.

Surjective: take arbitrary $b \in B$, *construct* an a and check $f(a) = b$.

Not surjective: exhibit one b and *prove* nothing maps to it. ■

WORKED: $F : \mathbb{R} \rightarrow \mathbb{R}, F(X) = 3X - 5$

Injective: $3a_1 - 5 = 3a_2 - 5 \Rightarrow 3a_1 = 3a_2 \Rightarrow a_1 = a_2$.

Surjective: given y , set $x = \frac{y+5}{3}$; then $f(x) = 3 \cdot \frac{y+5}{3} - 5 = y$. So f is a bijection.

Classify, then prove it

ACTIVITY 2 GROUPS OF THREE

15 MIN

For each: classify as injective, surjective, both or neither – then prove it with a template.

1. $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(n) = 2n + 1$

2. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3$

3. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$

4. $f : \mathbb{N} \rightarrow \mathbb{N}, f(n) = n + 1$

5. $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}, f(a, b) = a - b$

6. $f : \mathbb{R} \rightarrow (0, \infty), f(x) = 2^x$

Share: one group gives a *positive* proof, one a counterexample. The room votes first.

Six classifications, each with its reason

<i>F</i>	INJECTIVE?	SURJECTIVE?	THE REASON
$1 \cdot \mathbb{Z} \rightarrow \mathbb{Z}, 2n + 1$	Yes	No	Strictly increasing; but $2n + 1$ is odd, so 2 is missed.
$2 \cdot \mathbb{R} \rightarrow \mathbb{R}, x^3$	Yes	Yes	Bijective , inverse $\sqrt[3]{y}$.
$3 \cdot \mathbb{R} \rightarrow \mathbb{R}, x^2$	No	No	$f(-1) = f(1)$; and -1 is never attained.
$4 \cdot \mathbb{N} \rightarrow \mathbb{N}, n + 1$	Yes	No	Nothing maps to 1 (taking $\mathbb{N}$ to start at 1).
$5 \cdot \mathbb{Z}^2 \rightarrow \mathbb{Z}, a - b$	No	Yes	$f(1, 1) = f(2, 2)$; every k comes from $(k, 0)$.
$6 \cdot \mathbb{R} \rightarrow (0, \infty), 2^x$	Yes	Yes	Bijective , inverse $\log_2 y$.

WATCH OUT

"The function x^2 " names no function at all

DOMAIN	CODOMAIN	INJECTIVE?	SURJECTIVE?	SO IT IS...
$\mathbb{R}$	$\mathbb{R}$	No (± 2)	No (-1)	neither
$[0, \infty)$	$\mathbb{R}$	Yes	No (-1)	injective only
$\mathbb{R}$	$[0, \infty)$	No (± 2)	Yes	surjective only
$[0, \infty)$	$[0, \infty)$	Yes	Yes	bijjective

TRAP One rule, four functions, four different answers. A question asking "is x^2 injective?" is not yet a question.

FIX Write $f : A \rightarrow B$ **before** you write the rule. Every answer you write this term starts that way.

PART II

Composing, and undoing

Functions are things you can chain. Some chains can be reversed — and the condition for that is exactly the vocabulary from the first hour.



Composition needs the codomain of one to be the domain of the next

DEFINITION The **composition** of $f : A \rightarrow B$ with $g : B \rightarrow C$ is

$$(g \circ f)(a) = g(f(a)), \quad g \circ f : A \rightarrow C$$

Read it right to left: f runs first. If the middle sets do not match, the composition does not exist.

ORDER IS NOT A DETAIL

$f(x) = x + 1, g(x) = x^2$, both $\mathbb{R} \rightarrow \mathbb{R}$.

$$(g \circ f)(x) = (x + 1)^2 \quad (f \circ g)(x) = x^2 + 1$$

At $x = 1$: 4 against 2. So $g \circ f \neq f \circ g$.

Injectivity of a chain is inherited by its first link

THEOREM Let $f : A \rightarrow B$ and $g : B \rightarrow C$. If $g \circ f$ is injective, then f is injective.

PROOF Suppose $f(a_1) = f(a_2)$. Applying g to both sides gives $g(f(a_1)) = g(f(a_2))$, that is $(g \circ f)(a_1) = (g \circ f)(a_2)$. Since $g \circ f$ is injective, $a_1 = a_2$. Hence f is injective. ■

AND g ? Nothing follows. With $A = \{1\}$, $B = \{1, 2\}$, $C = \{1\}$, $f(1) = 1$, $g(1) = g(2) = 1$: the chain is injective, g is not.

A function can be undone exactly when it is a bijection

THEOREM $f : A \rightarrow B$ has a $g : B \rightarrow A$ with $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$ **if and only if** f is a bijection. This g is unique: it is f^{-1} .

PROOF ($\Leftarrow$) Let $b \in B$. Surjectivity gives an a with $f(a) = b$; injectivity says it is the only one. So $g(b) = a$ is well defined, and both composites are identities. ■

TRAP f^{-1} is not $\frac{1}{f}$. For $f(x) = 3x - 5$: $f^{-1}(1) = 2$, while $\frac{1}{f(1)} = -\frac{1}{2}$.

FIX f^{-1} undoes; $\frac{1}{f}$ divides. Same superscript, unrelated operations.

Find the broken step

ACTIVITY 3 GROUPS OF THREE

14 MIN

Claim. If $g \circ f$ is injective, then g is injective. **"Proof."**

- (i) Suppose $g(b_1) = g(b_2)$.
- (ii) Write $b_1 = f(a_1)$ and $b_2 = f(a_2)$.
- (iii) Then $(g \circ f)(a_1) = (g \circ f)(a_2)$, so $a_1 = a_2$.
- (iv) Hence $b_1 = f(a_1) = f(a_2) = b_2$. ■

1. Which of (i)–(iv) is the first unjustified line?
2. Give a counterexample showing the claim is false.
3. Add the smallest hypothesis on f that repairs the proof.
4. State and prove the companion result for $g \circ f$ *surjective*.

Share: one group presents the counterexample, another the repaired statement. The room decides if step 3 is the smallest fix.

Step (ii) assumed what was never given

1 · THE FIRST UNJUSTIFIED LINE IS (II) "Write $b_1 = f(a_1)$ " assumes every element of B is an output of f – that is surjectivity, never assumed.

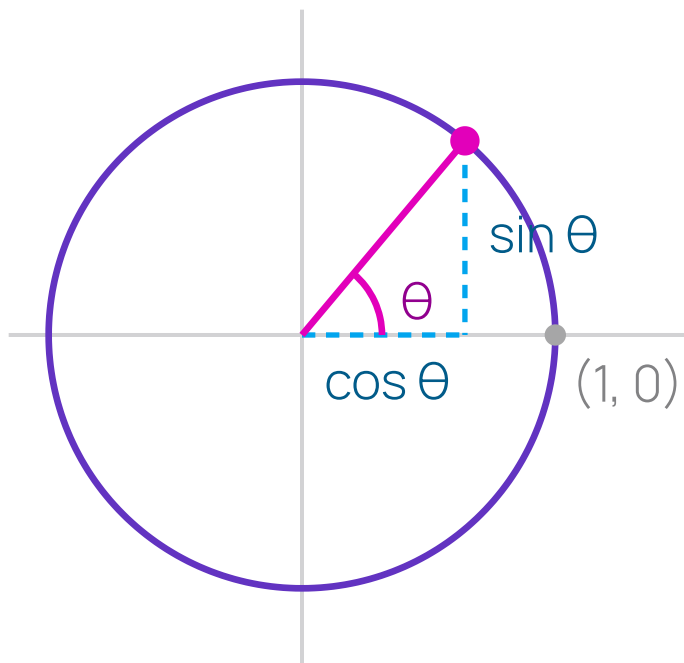
2 · COUNTEREXAMPLE $f : \{1\} \rightarrow \{1, 2\}$, $f(1) = 1$; and $g : \{1, 2\} \rightarrow \{1\}$, $g(1) = g(2) = 1$. Then $g \circ f$ is injective but g is not.

3 · THE REPAIR Assume f is **surjective**. Step (ii) is then legitimate and the rest stands.

4 · THE COMPANION RESULT If $g \circ f$ is surjective, so is g : given $c \in C$, pick a with $g(f(a)) = c$; then $f(a)$ is a preimage.



Sine and cosine are coordinates, and the identity is Pythagoras



DEFINITION From $(1, 0)$, walk θ counter-clockwise along the unit circle $x^2 + y^2 = 1$. The point you reach is $(\cos \theta, \sin \theta)$, and $\tan \theta = \frac{\sin \theta}{\cos \theta}$ where $\cos \theta \neq 0$. That distance is θ **radians**.

WHY THE IDENTITY IS FREE The point lies *on* the circle, so $x^2 + y^2 = 1$.
Substitute:

$$\cos^2 \theta + \sin^2 \theta = 1$$

Not a new fact — the circle's equation, renamed. ■

Periodicity gives repeated outputs

F	DOMAIN	IMAGE	PERIOD	INJECTIVE THERE?
$\sin$	$\mathbb{R}$	$[-1, 1]$	2π	No ($\sin 0 = \sin \pi$)
$\cos$	$\mathbb{R}$	$[-1, 1]$	2π	No ($\cos(-\theta) = \cos \theta$)
$\tan$	$\mathbb{R} \setminus \{\frac{\pi}{2} + k\pi\}$	$\mathbb{R}$	π	No (period π)

DEFINITION $p > 0$ is a **period** of f if $f(\theta + p) = f(\theta)$ for every θ . When there is a smallest positive one, it is the **fundamental period**.

THE RESTRICTION TRICK, AGAIN $\sin$ cut down to $[-\frac{\pi}{2}, \frac{\pi}{2}] \rightarrow [-1, 1]$ is a **bijection**, so by the theorem it has an inverse. That inverse is **arcsin**.

The unit circle definition already fixed your units

TRAP

$$\sin(30) \neq \frac{1}{2}.$$

θ is an arc length, so $\sin(30) \approx -0.988$: the point after walking 30 units round a circle of circumference 2π .
What you meant was $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$.

FIX

Convert once, at the start: $\theta_{\text{rad}} = \frac{\pi}{180} \theta_{\text{deg}}$.

Radians are the default here, in every calculus course, and in every programming language's standard library. Degrees are the special case that needs announcing.

THIRTY SECONDS Convert 45° , 60° and 270° to radians. Then: what is $\cos(180)$, and what did whoever wrote it mean?

Derive the circle without us

ACTIVITY 4 PAIRS

12 MIN

Draw one unit circle. Close every other source — no tables, no phones.

1. Place $\theta = \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}$ and read off $(\cos \theta, \sin \theta)$ from the 30–60–90 and 45–45–90 triangles.
2. From the picture alone, argue $\cos(-\theta) = \cos \theta$ and $\sin(-\theta) = -\sin \theta$.
3. From the picture alone, argue $\sin(\theta + \pi) = -\sin \theta$.
4. 2π is a period of **tan**. Is it the *smallest* one? Justify.

Share: one pair draws the circle and defends step 4 at the board.

Everything read off one picture

1 · THE THREE ANGLES

$$\frac{\pi}{6} : \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right) \cdot \frac{\pi}{4} : \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right) \cdot \frac{\pi}{3} : \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right)$$

2 · EVEN AND ODD

Negating θ reflects the point in the x -axis: the first coordinate is unchanged and the second flips. So $\cos(-\theta) = \cos \theta$, $\sin(-\theta) = -\sin \theta$. ■

3 · ADDING π

Adding π sends the point to its **antipode**, negating both coordinates. Reading off the second gives $\sin(\theta + \pi) = -\sin \theta$. ■

4 · "A PERIOD" IS NOT "THE PERIOD"

No. $\tan(\theta + \pi) = \tan \theta$, so π is smaller — and π is in fact the smallest. Verifying one value never establishes minimality.

On your own, before you leave

1 Choose a domain and codomain that make $x \mapsto x^2 - 4x$ a bijection, and prove both halves.

2 With $f(x) = 2x + 1$ and $g(x) = x^2$ on $\mathbb{R}$: compute $g \circ f$ and $f \circ g$, and give one x where they differ.

3 Derive $\cos \frac{\pi}{3}$ and $\sin \frac{\pi}{3}$ from the unit circle. Recalling them scores zero.

RULE FOR TODAY Every verdict carries its template. "Injective" alone is not an answer; "injective, because $f(a_1) = f(a_2)$ forces $a_1 = a_2$ since..." is.

What today was actually about

- A rule is not a function. **Domain and codomain** are part of the **object** — change them and every answer changes.
- Injective and surjective are **claims to be proved**, each with a fixed opening move.
- A function is invertible exactly when it is a bijection — which is why **arcsin** needs a restricted domain.

domain

codomain

image

preimage

bijjective

composition

inverse

radian

period

Problem Set 3

Eight injective/surjective classifications, each with a proof or a counterexample, plus five unit-circle values *derived*, not recalled.

Next session: **Function families** — polynomial, exp/log and trig, and how fast each one grows.